

Scratch: Move & Turn

A Beginner Coding Lesson for Kids

What you'll learn

- What Scratch is
- What sprites are
- How to use the Move block
- How to use the Turn blocks
- How to combine blocks

Who is this for?

- Kids aged 6–10
- Complete beginners
- Classroom or home use
- No prior coding needed

1. What is Scratch?

Scratch is a fun and easy coding tool made just for kids. It lets you tell stories and make games by giving instructions to characters called **sprites**. Instead of typing complicated code, you simply snap colourful blocks together — just like building with toy bricks!

Key Fact

Scratch was created at MIT and is free to use at scratch.mit.edu

2. Meet the Sprites!

Sprites are the characters or objects that move around on the **Stage** (the white area where your program runs). You can:

Option	How to do it
Choose from the library	Click the sprite icon and pick a cat, ball, butterfly, or more!
Upload your own image	Draw your character on paper, take a photo, and upload it.
Draw in Scratch	Use the built-in paint editor to create something totally unique!

3. The Move Block

The **Move block** tells your sprite to walk forward a certain number of steps. The Stage is 480 steps wide and 360 steps tall, so each 'step' is one pixel on screen.

Block	What it does
move 10 steps	Moves the sprite forward 10 pixels (small move)
move 50 steps	Moves the sprite forward 50 pixels (bigger move)

move -10 steps

Negative number = moves BACKWARD!

Remember:** More steps = bigger move!
Try changing the number to see how far your sprite goes.

4. The Turn Blocks

Turn blocks spin your sprite left or right by a number of **degrees**. A full circle = 360 degrees. There are two turn blocks:

Block	Direction	Example
turn clockwise 15°	RIGHT (clockwise)	Small 15° spin to the right
turn counter-clockwise 15°	LEFT (counter-clockwise)	Small 15° spin to the left
turn clockwise 90°	RIGHT (clockwise)	Quarter turn — sprite now faces down
turn clockwise 180°	RIGHT (clockwise)	Half turn — sprite faces the opposite way

Clockwise vs Counter-clockwise
Clockwise = turning RIGHT (like clock hands)
Counter-clockwise = turning LEFT (opposite to clock hands)

5. Combining Move & Turn!

The real magic happens when you use both blocks together. You can create any path you like — straight lines, corners, zigzags, even circles!

Code sequence	What happens
move 50 steps turn clockwise 90° move 50 steps	Sprite moves right, turns to face downward, then moves down — making an L-shape
turn clockwise 90° move 50 steps (repeat 4 times)	Sprite draws a perfect square on the stage!
turn clockwise 10° move 5 steps (repeat many times)	Sprite traces a circular path!

6. Let's Write Our First Code!

Follow these steps to make your sprite turn and move for the first time:

- 01 Open Scratch at scratch.mit.edu on your computer or tablet.
- 02 Click the yellow "Motion" category in the block palette on the left.
- 03 Drag the "move 20 steps" block into the coding area (middle panel).
- 04 Drag the "turn clockwise 90 degrees" block and snap it below the first block.
- 05 Click the green flag (■) at the top of the Stage to run your code.
- 06 Watch your sprite turn to the right and walk forward!

7. Fun Practice Activities

Move Forward & Backward

Use the Move block to make your sprite walk forward. Then change the number to a negative value (e.g. -30) to make it go backward!

Turn Left & Right

Try turning your sprite with different degree values: 15°, 45°, and 90°. Notice how the bigger the number, the more it spins!

Draw a Square!

Move 50 steps, then turn 90° — repeat this 4 times. Your sprite will trace a perfect square on the stage!

Make a Circle!

Turn 10° clockwise then move 5 steps — repeat this many times. Your sprite will gradually trace a circular path!

Mix & Experiment!

Combine Move and Turn blocks in any order you like. There are no wrong answers — every try teaches you something new!

8. Tips for Scratch Success

- ◆ **Take Your Time:** Drag blocks slowly and carefully. Coding is about thinking and trying — there's no rush!
- ◆ **Use Small Numbers First:** Try moving just 10 steps or turning only 15° to see what happens before going bigger.
- ◆ **Experiment Freely:** Change numbers in your blocks and watch what your sprite does. Every try teaches you something new!
- ◆ **Ask for Help:** If you get stuck, ask a friend, your teacher, or a parent. Coders always help each other!

9. Quick Reference Card

Block	What it does	Example
move N steps	Walk forward N pixels	move 50 steps
move -N steps	Walk backward N pixels	move -20 steps
turn CW N degrees	Spin right by N degrees	turn CW 90 degrees
turn CCW N degrees	Spin left by N degrees	turn CCW 45 degrees

Every coder starts with a single block — Keep exploring, keep coding!